

Uditraj Singh Rathore

GitHub Link: <https://github.com/uditraj1> | Portfolio Website: [Uditraj Singh Rathore Portfolio \(uditraj1.github.io\)](https://uditraj1.github.io)

Washington, DC, 20009 | (240)936-3470 | uditrajsinghrathore50@gmail.com | [LinkedIn Profile](#)

EDUCATION

University of Maryland

M.Eng of Software Engineering

SRM Institute of Science and Technology

B.Tech, Computer Science and Engineering

College Park, MD, USA

August 2022 - May 2024

Chennai, TN, India

July 2016 - May 2020

PROFESSIONAL EXPERIENCE

System Automation Corporation

Solution Developer

Orlando, FL, USA

July 2024 – Present

- Utilized Python to develop automation scripts, backend services, and data processing tools, significantly reducing manual workloads, improving process efficiency, and enhancing overall system reliability by 50%
- Built custom React widgets integrated with Redux for efficient state management, including features like the GIS maps widget and interactive dashboards components and import widget, enhancing user experience and increasing user engagement by 30%
- Designed and implemented scalable system architecture, ensuring reliability, security, and maintainability across distributed environments, resulting in improving application performance by 45%
- Leveraged Terraform to provision and manage cloud infrastructure as code, automating deployments, enhancing scalability, improving consistency, and reducing configuration errors across multiple environments by 40%

Fraunhofer USA Center Mid-Atlantic (CMA)

Web and ML Engineer

Riverdale Park, MD, USA

January 2023 – May 2024

- Implemented and optimized high-performance backend services using Golang, implementing efficient concurrency patterns and scalable APIs, improving system throughput, reducing latency, and enhancing reliability by 35%
- Developed a computer vision object detection tool using Python that was tested by another AI, resulting in improving the application accuracy compared to previous machine learning models by 25%
- Designed and developed large language models (LLMs), optimizing training pipelines, fine-tuning model parameters, resulting in more context-aware language outputs, enhancing natural language understanding accuracy by 40%
- Implemented Datadog to monitor and analyze ML system performance, tracking model latency, resource utilization, and inference accuracy, improving observability and enhancing model reliability by 50%

Ford Motor Company

Software Engineer

Chennai, TN, India

July 2020 – August 2022

- Programmed in Angular and Typescript to develop dynamic and responsive user interfaces for the front-end development of the system, resulting in enhancing interactivity and user experience by 45%
- Created and optimized REST APIs to enable seamless communication between distributed services, improving data exchange, enhancing system interoperability and boosting overall application performance by 55%
- Implemented multiple software applications using the Spring Boot framework, leveraging Java and MSSQL for backend, advancing software efficiency and server response times by 35%
- Leveraged design patterns such as MVC and Observer in developing complex web applications, ensuring modularity and reusability, improving code maintainability, scalability, and overall system performance by 30%

Vinfotech

Web Development Intern

Indore, MP, India

June 2017 – July 2017

- Created robust web applications using JavaScript frameworks, resulting in an enhancement of customer experience and functionality by 20%

TECHNICAL SKILLS

Programming/Technical Skills: C, C++, Java, Javascript, Typescript, HTML/HTML5, Python, CSS3, RESTful APIs, SOAP API, Design Reviews, Code Reviews, Data Structures, Algorithms, System Design, Frontend Development, TCP, IP, UI/UX, Docker, Backend Development, Feature Development, Database, Full-Stack Development, Object-Oriented Designs, Microservices, Technical Documentatiton, Kubernetes, CI/CD Pipelines, Software Development Life Cycle (SDLC), Machine Learning (ML), AI, LLM, MySQL, Microsoft SQL Server, PostgreSQL, DynamoDB, MongoDB, NoSQL, Redis Caching

Additional Skills/Soft Skills: VS Code, Git, Jenkins, AngularJS, ReactJS, Next.js, Google Cloud Platform (GCP), JUnit, Unit Testing, Amazon Web Services (AWS), ECS, S3, Microsoft Azure, Spring Boot, End-To-End Testing, Version Control, Agile, Scrum, Debugging, Compliance, Problem-Solving, Ownership Mindset, Attention To Detail, Leader, Accountability, Self-Motivated, Team-Oriented, Communication Skills, Curiosity For Learning, Collaborative Attitude

PERSONAL PROJECTS

- E-LMS on React, Java and Spring Boot (Oct 2023):** React & Spring Boot e-learning platform: signup, login, admin controls, email verification, enrollment, dashboards material access, grayscale mode, & instructor features